

Fixed Points IV: Curvature, Centroid Motion, and Structural Transitions

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Part IV of VI in the Fixed Point series. As both the cornerstone of the Modal Triplet Theory (MTT) collection and a stand-alone development, the series is intended to function simultaneously as a basis and as a self-contained study. Each paper in the series builds upon its predecessors, extending the fixed-point framework step by step.

Abstract

We rebuild curvature coupling and structural transitions on the corrected FP-I-III spine. A Laplace-type operator is normalized as $L = \nabla^* \nabla + \mathcal{R}$. Curvature can shift the spectral cluster, mix coherent and noncoherent sectors, and create a leakage floor. If $\|\mathcal{R}\| < \lambda_*/2$, the full curved operator retains a separated low cluster and defines a new Riesz projector; if one retains the unperturbed projector, the term QRP must appear explicitly. Centroids are defined intrinsically by a Karcher mean inside a convex normal ball. A first-order gradient parent flow yields a first-order modulation law; a Newtonian law requires a separately specified inertial parent equation. Absolute interaction bounds give $|E_{\text{int}}| \leq CO$, while attraction or repulsion needs a sign hypothesis. Structural transitions are controlled by a Lyapunov mountain- pass/work theorem, and exit detection is separated from selection of the post-transition basin. We also state the exact projective-module transport and finite-reduction boundary: a supplied connection and its Hessian transport naturally, bare compression is exact only on an invariant subspace, and otherwise the Feshbach or reduced-Green operator is required.

Revision note for this edition

Supersedes. *Fixed Points IV: Curvature, Centroid Motion, and Structural Transitions*, version 4.

Reason. Version 4 corrected curvature leakage and transition claims but did not state when a curved continuum Hessian is represented exactly by finite projective-module data or when a compressed matrix is the physical effective operator.

Resolution. Version 5 retains the curved Riesz-projector analysis and adds exact connection/Hessian transport, the invariant-subspace versus Feshbach–Schur criterion, and the reduced-Green shorted Hessian for nonlinear strain coordinates. It also distinguishes a finite matrix-valued smooth projector from a Fourier or Galerkin truncation.

Retained result. Curved-cluster persistence, leakage bounds, intrinsic first-order modulation, and barrier exclusion survive conditionally.

Remaining boundary. The selected q79 HYM endpoints, action, physical Hessian and finite subspace remain open, as do attraction, inertia, and post-transition selection without a specified physical parent law.

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1 Inherited control framework

We use the corrected FP-I/II Riemannian control geometry, joint internal operators, and projected stabilization flow. The projector P below is the unperturbed joint harmonic projector and $Q = I - P$. FP-III supplies separate deterministic and stochastic disturbance bounds. No statement here identifies the stabilization parameter with physical Lorentzian time.

Existence of a projected step fixed point requires the corrected invariant-set conditions and the stated Schauder or Darbo hypotheses. Promotion to a full equilibrium uses the strict Lyapunov identity of FP-II.

2 Curved Laplace-type operator

Let $E \rightarrow B$ be a Hermitian bundle with compatible connection ∇ . A Laplace-type operator is written

$$L = \nabla^* \nabla + \mathcal{R}, \quad (2.1)$$

where $\nabla^* \nabla \geq 0$ is the rough/connection Laplacian and $\mathcal{R} = \mathcal{R}^*$ is the curvature endomorphism. We identify the unperturbed vertical operator with $A_0 = \nabla^* \nabla$ in the selected sector and assume

$$\text{spec}(A_0) \subset \{0\} \cup [\lambda_*, \infty), \quad \lambda_* > 0.$$

Relative to $P \oplus Q$, curvature has four blocks

$$\mathcal{R} = \begin{pmatrix} PRP & PRQ \\ QRP & QRQ \end{pmatrix}.$$

Define the noncoherent negative-part and leakage bounds

$$\operatorname{Re}\langle q, Q\mathcal{R}Qq \rangle \geq -\rho_Q \|q\|^2, \quad (2.2)$$

$$\ell_{QP} := \|Q\mathcal{R}P\|. \quad (2.3)$$

The scalar lower bound ρ_Q does not control the off-diagonal block $Q\mathcal{R}P$.

3 Gap persistence and the curved projector

Theorem 3.1 (Bounded-curvature cluster persistence). *Assume $\mathcal{R}$ is bounded and self-adjoint with $r := \|\mathcal{R}\| < \lambda_*/2$. Then the spectrum of $L = A_0 + \mathcal{R}$ lies in the r -neighborhood of $\operatorname{spec}(A_0)$. The low cluster and the remaining spectrum are separated by at least*

$$\lambda_{\text{gap}}^{\mathcal{R}} \geq \lambda_* - 2r > 0.$$

A contour in this separation defines the full curved Riesz projector

$$P_{\mathcal{R}} = \frac{1}{2\pi i} \int_{\Gamma} (z - L)^{-1} dz,$$

which has the same finite rank as P along the norm-continuous perturbation $A_0 + s\mathcal{R}$, $0 \leq s \leq 1$.

Using $P_{\mathcal{R}}$ eliminates linear leakage because $[L, P_{\mathcal{R}}] = 0$, but changes the coherent subspace. If the bounded- perturbation condition is unavailable, gap persistence and the Riesz projector must be proved directly for the full curved operator; the unperturbed gap may not simply be reused.

For base-dependent curvature and projectors, modulation and locality estimates also require bounds on $\nabla\mathcal{R}$, $\nabla P_{\mathcal{R}}$, and any commutator with the base control operator. These gradient terms are not contained in the scalar loss ρ_Q .

4 Exact projective transport and finite reduction

The curved spectral projector above should not be confused with a finite projective-module presentation of a supplied bundle. Let X be compact and let (E, h, ∇) be a supplied finite-rank Hermitian bundle with unitary connection. The universal-connection construction gives a finite N , a fiberwise isometry

$$U : E \longrightarrow X \times \mathbb{C}^N, \quad p_{\text{mod}} = UU^* \in M_N(C^\infty(X)),$$

such that

$$p_{\text{mod}} d(Us) = U \nabla s, \quad p_{\text{mod}} (dp_{\text{mod}}) \wedge (dp_{\text{mod}}) p_{\text{mod}} = U F_{\nabla} U^*.$$

Consequently, a coupled differential and any gauge-fixed Hessian constructed functorially from the same supplied connection, metric, gauge slice, and action transport exactly under U [6]. This preserves nonzero Chern data; it does not flatten the connection.

Here “finite” refers only to the ambient matrix size N . The entries of $p_{\text{mod}}(x)$ are smooth functions and can contain infinitely many base modes. Thus p_{mod} is not, by itself, a finite Fourier, Toeplitz, or Galerkin cutoff. Given endpoint Cech data, metrics, a finite good cover, partition of unity, and local physical connection forms, the verified connection compiler emits the projector together with its connection correction. The projector alone carries only the partition-induced Grassmann connection and is not generally the physical HYM connection. Neither construction selects the physical q79 endpoints.

Theorem 4.1 (Exact finite-reduction criterion). *Let $H = H^*$ be the transported physical Hessian, let P_f be a finite orthogonal projector preserving its declared domain, and put $Q_f = I - P_f$. Then the bare compression $P_f H P_f$ is the exact restriction of H to $\text{Ran } P_f$ if and only if*

$$Q_f H P_f = 0.$$

If this residual is nonzero and $Q_f(H - z)Q_f$ is invertible, the exact finite operator at spectral parameter z is the Feshbach–Schur map

$$F_{P_f}(H - z) = P_f(H - z)P_f - P_f H Q_f [Q_f(H - z)Q_f]^{-1} Q_f H P_f.$$

The second term is a same-source complementary-sector self-energy, not an independent fit parameter.

Theorem 4.2 (Reduced-Green strain Hessian). *Let $H_Q > 0$ be the gauge-fixed physical Hessian after removal of its declared harmonic kernel, let $G_Q = H_Q^{-1}$, and let $J = d\Phi_{\text{strain}}$ be the differential of a declared nonlinear strain symbol at a regular point. If J has full target rank and $JG_Q J^* > 0$, then the exact effective Hessian on strain variations is*

$$H_{\text{strain}} = (JG_Q J^*)^{-1}.$$

Indeed, for every target variation y ,

$$\min_{Jx=y} \frac{1}{2} \langle x, H_Q x \rangle = \frac{1}{2} \langle y, (JG_Q J^*)^{-1} y \rangle.$$

Thus the effective operator integrates out the complementary physical fields and forgotten phase/orientation directions. It is a quotient or shorted Hessian, not the restriction of H_Q to a linear rank-six subspace.

For the current q79 program these are closed universal theorems and exact finite witnesses. Their physical instantiation remains open until the visible/hidden HYM endpoints, action-derived Hessian, regular quotient domain, and intrinsic finite subspace are supplied and either $Q_f H P_f = 0$ or the displayed Feshbach equality is executed.

5 Curvature leakage and stability floor

Consider

$$\partial_t \Psi = -(A_0 + \mathcal{R})\Psi - N(\Psi) + F, \quad p = P\Psi, \quad q = Q\Psi.$$

Assume unperturbed coherence invariance $QN(p) = 0$ and the one-sided estimate

$$\text{Re} \langle q, Q(N(p + q) - N(p)) \rangle \geq -L_Q \|q\|^2.$$

Theorem 5.1 (Noncoherent leakage bound). *Let*

$$\gamma_Q := \lambda_* - L_Q - \rho_Q > 0.$$

Then

$$D^+ \|q(t)\| \leq -\gamma_Q \|q(t)\| + \ell_{QP} \|p(t)\| + \|QF(t)\|. \quad (5.1)$$

Consequently,

$$\|q(t)\| \leq e^{-\gamma_Q t} \|q(0)\| \quad (5.2)$$

$$+ \int_0^t e^{-\gamma_Q(t-s)} (\ell_{QP} \|p(s)\| + \|QF(s)\|) ds. \quad (5.3)$$

If p and QF are uniformly bounded, then

$$\limsup_{t \rightarrow \infty} \|q(t)\| \leq \frac{\ell_{QP} \sup_t \|p(t)\| + \sup_t \|QF(t)\|}{\gamma_Q}.$$

Thus positive damping need not drive the old Q component to zero. Exact decay to the old coherent sector requires $Q\mathcal{R}P = 0$ and vanishing noncoherent forcing, or replacement of P by a verified curved projector.

6 Intrinsic centroid and modulation

Let

$$d\nu_t(y) = \frac{\rho(y, t) d\text{vol}_Y(y)}{\int_Y \rho(\cdot, t) d\text{vol}_Y}$$

be the normalized density measure. Assume its support lies in a geodesically convex normal ball on which the squared-distance functional is strictly convex.

Definition 6.1 (Karcher centroid). The centroid is the unique minimizer

$$X(t) = \operatorname{argmin}_{z \in Y} \frac{1}{2} \int_Y d_Y(z, y)^2 d\nu_t(y).$$

This definition is coordinate invariant. A coordinate average is used only after choosing a normal chart and estimating its curvature-dependent error.

Let $\{\Psi_X : X \in \mathcal{M}\}$ be a smooth localized profile manifold and write

$$\Psi(t) = \Psi_{X(t)} + r(t), \quad \langle r, \partial_{X^i} \Psi_X \rangle = 0.$$

Define the positive tangent Gram matrix $G_{ij}(X) = \langle \partial_{X^i} \Psi_X, \partial_{X^j} \Psi_X \rangle$ and $V_{\text{eff}}(X) = C[\Psi_X]$.

Theorem 6.2 (First-order modulation law). *For the first-order gradient flow $\partial_t \Psi = -\nabla C(\Psi)$, assume the profile decomposition is unique, $G(X)$ is uniformly invertible, and the remainder and curvature-gradient terms are controlled. Projection onto the tangent space gives*

$$G_{ij}(X) \dot{X}^j = -\partial_i V_{\text{eff}}(X) + \varepsilon_i(t),$$

where ε_i is bounded by the modulation remainder, localization width, $\nabla \mathcal{R}$, and projector-variation errors specified by the model.

A second-order law $M_{ij}(X) \ddot{X}^j + \dots = -\partial_i V_{\text{eff}}$ is retained only when the parent field equation contains an explicit inertial/kinetic second-order term. It does not follow from the FP first-order gradient flow.

7 Overlap and interaction sign

For two profiles define

$$E_{\text{int}} := C[\Psi_1 + \Psi_2] - C[\Psi_1] - C[\Psi_2].$$

Let $\mathcal{O}(\Psi_1, \Psi_2) \geq 0$ be a declared overlap functional.

Assumption 7.1 (Absolute cross-term control). All quadratic and nonlinear cross terms obey

$$|E_{\text{int}}| \leq C_{\text{int}} \mathcal{O}(\Psi_1, \Psi_2).$$

This assumption yields exactly the displayed absolute estimate. It does not yield positive constants c_1, c_2 with $c_1 \mathcal{O} \leq E_{\text{int}} \leq c_2 \mathcal{O}$.

Theorem 7.2 (Signed interaction criterion). *If a model additionally proves $E_{\text{int}} \leq -c\mathcal{O}$ with $c > 0$, overlap lowers the energy and is energetically attractive. If it proves $E_{\text{int}} \geq c\mathcal{O}$, overlap raises the energy and is energetically repulsive. Neither sign follows from overlap magnitude alone.*

Energetic attraction still does not prove dynamical merger. The flow must have access to the merged basin and avoid intervening conserved quantities or barriers.

8 Existence and equilibrium status

The corrected FP–I/II theorems apply to the curved flow only after the full operator supplies the required sectoriality, smoothing, invariant set, compactness/confinement, and projector hypotheses. A Schauder or Darbo result first gives a projected step fixed point. It becomes an equilibrium only when the strict Lyapunov identity is verified. Curvature does not preserve the old fixed-point structure automatically.

9 Structural transitions

Let $\mathcal{B}_-$ and $\mathcal{B}_+$ be two basins for a continuous gradient flow with Lyapunov energy C . Define the mountain-pass level

$$c_{-+} := \inf_{\gamma \in \Gamma_{-+}} \max_{s \in [0,1]} C(\gamma(s)),$$

where Γ_{-+} is the set of continuous paths joining the two basins.

Theorem 9.1 (Energy-barrier exclusion with work). *Assume trajectories are continuous in the energy topology and satisfy*

$$C[\Psi(t)] \leq C[\Psi(0)] + W_{\text{ext}}(t), \quad W_{\text{ext}}(t) \leq W_*.$$

If $C[\Psi(0)] + W_ < c_{-+}$, the trajectory cannot pass from $\mathcal{B}_-$ to $\mathcal{B}_+$. For an unforced gradient flow, $W_* = 0$.*

Noise-driven transitions require probabilistic exit estimates and are not covered by this deterministic theorem.

Theorem 9.2 (Exit does not determine selection). *Let $\mathcal{A}$ be an admissible region and $\tau_{\text{exit}} = \inf\{t : \Psi(t) \notin \mathcal{A}\}$. Finite τ_{exit} detects loss of admissibility only. Identification of the post-exit basin requires a separate basin-accessibility and convergence or selection theorem.*

10 Conclusion

Curvature is not merely a scalar subtraction from a damping margin. It can shift the spectral cluster, rotate the coherent projector, and leak coherent amplitude into the old noncoherent sector. FP IV v5 makes those effects explicit and adds the exact boundary between natural

projective transport, bare compression, Feshbach reduction, and nonlinear shorting. Centroid motion is intrinsic and first order for a gradient parent flow; interaction signs and dynamic merger require additional hypotheses; and barrier crossing is controlled by total energy plus work. These corrected statements preserve the useful curvature and transition program without claiming that a finite projector, overlap, or positive damping alone selects a physical basin.

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